

Normalization and Batch Correction

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```
# Load necessary packages
library(tidyverse)
library(cowplot)
library(DESeq2)
library(sva)
library(ggrepel)

# TIFF output directory (created automatically)
tiff_dir <- "../results/TIFFs/2_DataNormalization"
dir.create(tiff_dir, showWarnings = FALSE, recursive = TRUE)
```

1. Import files

We load the workspace created in the previous exploratory analysis.

```
load(file = "../results/dds.RData")
```

We extract the batch information from the metadata. This is important to find and fix technical effects that are not related to the biology of the experiment.

```
head(metadata)
```

```
##      Condition Replicates batch
## C1      Control      C1_R1     1
## C2      Control      C1_R2     2
## T1 Treatment      T1_R1     1
## T2 Treatment      T1_R2     2
```

```
batch <- metadata$batch
batch
```

```
## [1] 1 2 1 2
```

We check that the filtered counts object is still a matrix, which is needed for the correction functions.

```
class(filtcounts)
```

```
## [1] "matrix" "array"
```

2. Batch Effect Correction

We use `ComBat_seq` to adjust the filtered raw counts. Unlike other tools, `ComBat_seq` keeps the counts as whole numbers, which is required by `DESeq2`.

```
adjcounts <- ComBat_seq(filtcounts,      # Filtered counts matrix
                        batch = batch,    # Batch variable
                        group = NULL)     # No biological variables for now
```

```
## Found 2 batches
## Using null model in ComBat-seq.
## Adjusting for 0 covariate(s) or covariate level(s)
## Estimating dispersions
## Fitting the GLM model
## Shrinkage off - using GLM estimates for parameters
## Adjusting the data
```

```
head(adjcounts)
```

```
##                               C1    C2    T1    T2
## 1c1|NZ_JBCFOM010000001.1_cds_AAGW04_RS00515_104    265    267    335    285
## 1c1|NZ_JBCFOM010000001.1_cds_AAGW04_RS01695_335   3857   3912   2725   2314
## 1c1|NZ_JBCFOM010000001.1_cds_AAGW04_RS01865_369   1432   1451   1784   1520
## 1c1|NZ_JBCFOM010000001.1_cds_WP_002208627.1_289    466    472    567    483
## 1c1|NZ_JBCFOM010000001.1_cds_WP_005971891.1_35   16684  16982  7564   6428
## 1c1|NZ_JBCFOM010000001.1_cds_WP_005973933.1_458   5661   5710  4519   3827
```

We save the adjusted counts matrix.

```
write.csv(adjcounts, file = "../results/adjcounts.csv", row.names = TRUE)
```

We update the `DESeqDataSet` object using the new batch-corrected matrix.

```
dds_adj <- DESeqDataSetFromMatrix(countData = adjcounts,
                                   colData = metadatos,
                                   design = ~ Condition)
```

```
## converting counts to integer mode
```

```
## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

```
dds_adj
```

```
## class: DESeqDataSet
## dim: 4566 4
## metadata(1): version
## assays(1): counts
## rownames(4566): lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS00515_104
##   lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01695_335 ...
##   lcl|NZ_JBCFOM010000085.1_cds_WP_342414474.1_4391
##   lcl|NZ_JBCFOM010000086.1_cds_WP_342414475.1_4392
## rowData names(0):
## colnames(4): C1 C2 T1 T2
## colData names(3): Condition Replicates batch
```

3. Exploratory analysis of adjusted counts

We apply a new Variance Stabilizing Transformation (VST) on the corrected data to see how well ComBat_seq worked using a PCA plot.

```
vsd2 <- vst(dds_adj, blind = FALSE)
pcaData2 <- plotPCA(vsd2, intgroup = "Condition", returnData = TRUE)
```

```
## using ntop=500 top features by variance
```

```
# Calculate explained variance
percentVar2 <- round(100 * attr(pcaData2, "percentVar"))
paste("Explained variance (PC1 + PC2) = ", sum(percentVar2), "%")
```

```
## [1] "Explained variance (PC1 + PC2) = 100 %"
```

We create the PCA plot for the adjusted data.

```
# Same palette and shapes as PCA in script 1
pca_palette <- c("Control" = "#4575B4", "Treatment" = "#D73027")

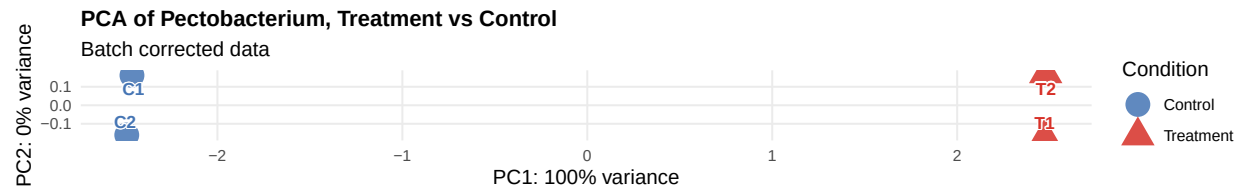
pca2 <- ggplot(pcaData2, aes(x = PC1, y = PC2,
                             color = Condition, shape = Condition)) +
  geom_point(size = 7, alpha = 0.85) +
  scale_shape_manual(values = c("Control" = 16, "Treatment" = 17),
                     name = "Condition") +
  scale_color_manual(values = pca_palette, name = "Condition") +
  geom_text_repel(aes(label = name), show.legend = FALSE,
                  size = 4, fontface = "bold",
                  box.padding = 0.5, bg.color = "white", bg.r = 0.1) +
  labs(title = "PCA of Pectobacterium, Treatment vs Control",
       subtitle = "Batch corrected data",
       x = paste0("PC1: ", percentVar2[1], "% variance"),
       y = paste0("PC2: ", percentVar2[2], "% variance")) +
```

```

theme_minimal(base_size = 13) +
theme(panel.grid.minor = element_blank(),
      plot.title = element_text(face = "bold", size = 14)) +
coord_fixed()

```

pca2



```

ggsave(filename = "../results/pca_adj.pdf", plot = pca2, width = 11, height = 8.5)
ggsave(filename = file.path(tiff_dir, "pca_adj.tiff"), plot = pca2, width = 11, height = 8.5, dpi = 300)

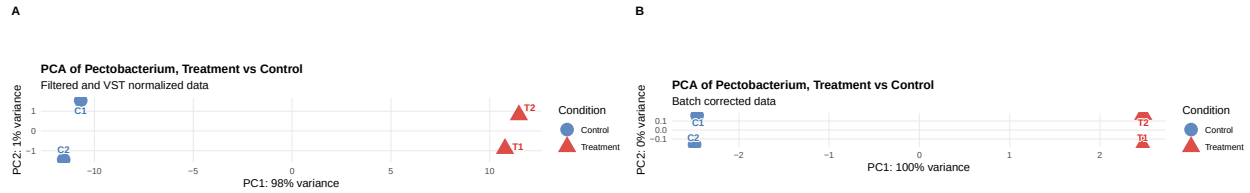
```

We visually compare the original (raw) PCA with the corrected (adjusted) PCA.

```

pcas <- plot_grid(pca, pca2,
                  ncol = 2, nrow = 1,
                  labels = "AUTO")
pcas

```



```
ggsave(filename = "../results/pca_raw_adj.pdf", plot = pcas, width = 20, height = 4)
ggsave(filename = file.path(tiff_dir, "pca_raw_adj.tiff"), plot = pcas, width = 20, height = 4, dpi = 300)
```

We save the progress of the adjusted data.

```
save(dds_adj, adjcounts, pca2, pcas, file = "../results/dds_adj.RData")
```

4. Normalization for Differential Expression

DESeq2 uses the “median of ratios” method to calculate size factors. This fixes differences in sequencing depth between samples.

```
dds_norm <- DESeq(dds_adj)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

```
dds_norm
```

```
## class: DESeqDataSet
## dim: 4566 4
## metadata(1): version
## assays(4): counts mu H cooks
## rownames(4566): lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS00515_104
##   lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01695_335 ...
##   lcl|NZ_JBCFOM010000085.1_cds_WP_342414474.1_4391
##   lcl|NZ_JBCFOM010000086.1_cds_WP_342414475.1_4392
## rowData names(22): baseMean baseVar ... deviance maxCooks
## colnames(4): C1 C2 T1 T2
## colData names(4): Condition Replicates batch sizeFactor
```

Review of the size factors for each sample: * Values > 1 mean the sample was over-represented. * Values < 1 mean the sample was under-represented.

```
dds_norm$sizeFactor
```

```
##           C1           C2           T1           T2
## 1.0482538 1.0501014 1.0302766 0.8823857
```

We extract and save the final normalized counts matrix.

```
normcounts <- counts(dds_norm, normalized = TRUE)
head(normcounts)
```

```
##
##          C1          C2          T1
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS00515_104 252.8014 254.2612 325.1554
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01695_335 3679.4523 3725.3545 2644.9208
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01865_369 1366.0813 1381.7713 1731.5738
## lcl|NZ_JBCFOM010000001.1_cds_WP_002208627.1_289 444.5488 449.4804 550.3376
## lcl|NZ_JBCFOM010000001.1_cds_WP_005971891.1_35 15915.9924 16171.7716 7341.7178
## lcl|NZ_JBCFOM010000001.1_cds_WP_005973933.1_458 5400.4096 5437.5701 4386.2008
##
##          T2
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS00515_104 322.9880
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01695_335 2622.4358
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01865_369 1722.6026
## lcl|NZ_JBCFOM010000001.1_cds_WP_002208627.1_289 547.3796
## lcl|NZ_JBCFOM010000001.1_cds_WP_005971891.1_35 7284.7958
## lcl|NZ_JBCFOM010000001.1_cds_WP_005973933.1_458 4337.1054
```

```
write.csv(normcounts, file = "../results/normcounts.csv", row.names = TRUE)
```

4.1. Evaluate data spread

To see the effect of our processing, we compare the counts in three stages: raw, batch-corrected, and normalized. We use a log transformation ($\log(x+1)$) just to make the boxplots easier to read.

```
log_raw <- log(filtcounts + 1)
log_adj <- log(adjcounts + 1)
log_norm <- log(normcounts + 1)
```

Helper function to change the matrices from wide format to long format. This makes them work with ggplot2.

```
wide2long <- function(countsmatrix) {
  as.data.frame(countsmatrix) %>%
  rownames_to_column("Gene") %>%
  pivot_longer(cols = -Gene,
               names_to = "Sample",
               values_to = "Count")
}
```

We apply this function to our three matrices:

```
log_raw_long <- wide2long(log_raw)
log_adj_long <- wide2long(log_adj)
log_norm_long <- wide2long(log_norm)
```

Create comparative boxplots:

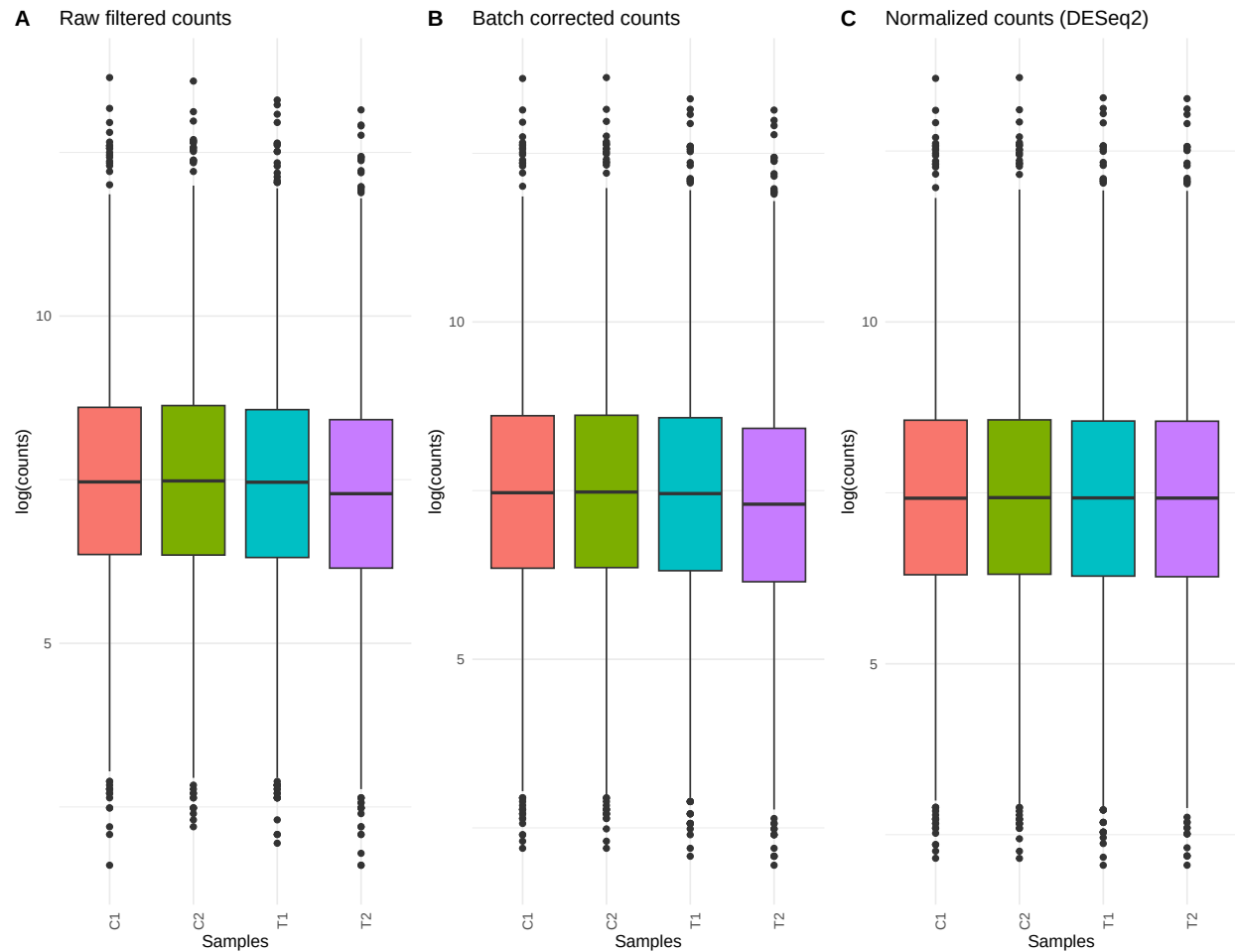
```
# 1. Raw counts boxplot
box1 <- ggplot(log_raw_long, aes(x = Sample, y = Count, fill = Sample)) +
  geom_boxplot() +
  labs(title = "Raw filtered counts",
       x = "Samples",
       y = "log(counts)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1),
        legend.position = "none")

# 2. Batch corrected boxplot
box2 <- ggplot(log_adj_long, aes(x = Sample, y = Count, fill = Sample)) +
  geom_boxplot() +
  labs(title = "Batch corrected counts",
       x = "Samples",
       y = "log(counts)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1),
        legend.position = "none")

# 3. Normalized counts boxplot
box3 <- ggplot(log_norm_long, aes(x = Sample, y = Count, fill = Sample)) +
  geom_boxplot() +
  labs(title = "Normalized counts (DESeq2)",
       x = "Samples",
       y = "log(counts)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1),
        legend.position = "none")
```

We group and save the final plots.

```
boxplots <- plot_grid(box1, box2, box3,
                      ncol = 3, nrow = 1,
                      labels = "AUTO")
boxplots
```



```
ggsave(filename = "../results/boxplots.pdf", plot = boxplots, width = 15, height = 6)
ggsave(filename = file.path(tiff_dir, "boxplots.tiff"), plot = boxplots, width = 15, height = 6, dpi = 300)
```

We save the final normalized objects for the differential expression analysis.

```
save(dds_norm, normcounts, file = "../results/dds_norm.RData")
```

Session information

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26120)
##
## Matrix products: default
## LAPACK version 3.12.1
```

```

##
## locale:
## [1] LC_COLLATE=Spanish_Mexico.utf8 LC_CTYPE=Spanish_Mexico.utf8
## [3] LC_MONETARY=Spanish_Mexico.utf8 LC_NUMERIC=C
## [5] LC_TIME=Spanish_Mexico.utf8
##
## time zone: America/Mexico_City
## tzcode source: internal
##
## attached base packages:
## [1] stats4      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] ggrepel_0.9.8          sva_3.58.0
## [3] BiocParallel_1.44.0    genefilter_1.92.0
## [5] mgcv_1.9-3             nlme_3.1-168
## [7] DESeq2_1.50.2          SummarizedExperiment_1.40.0
## [9] Biobase_2.70.0         MatrixGenerics_1.22.0
## [11] matrixStats_1.5.0      GenomicRanges_1.62.1
## [13] Seqinfo_1.0.0          IRanges_2.44.0
## [15] S4Vectors_0.48.1       BiocGenerics_0.56.0
## [17] generics_0.1.4         cowplot_1.2.0
## [19] lubridate_1.9.5        forcats_1.0.1
## [21] stringr_1.6.0          dplyr_1.2.1
## [23] purrr_1.2.2            readr_2.2.0
## [25] tidyr_1.3.2            tibble_3.3.1
## [27] ggplot2_4.0.3          tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1      farver_2.1.2          blob_1.3.0
## [4] Biostrings_2.78.0     S7_0.2.2              fastmap_1.2.0
## [7] XML_3.99-0.23         digest_0.6.39         timechange_0.4.0
## [10] lifecycle_1.0.5      statmod_1.5.1         survival_3.8-3
## [13] KEGGREST_1.50.0       RSQLite_3.52.0        magrittr_2.0.5
## [16] compiler_4.5.2        rlang_1.2.0           tools_4.5.2
## [19] yaml_2.3.12           knitr_1.51            labeling_0.4.3
## [22] S4Arrays_1.10.1      bit_4.6.0             DelayedArray_0.36.1
## [25] RColorBrewer_1.1-3    abind_1.4-8           withr_3.0.2
## [28] grid_4.5.2           xtable_1.8-8          edgeR_4.8.2
## [31] scales_1.4.0         cli_3.6.5             crayon_1.5.3
## [34] rmarkdown_2.31       ragg_1.5.2            otel_0.2.0
## [37] rstudioapi_0.18.0    httr_1.4.8           tzdb_0.5.0
## [40] DBI_1.3.0            cachem_1.1.0          splines_4.5.2
## [43] parallel_4.5.2       AnnotationDbi_1.72.0  XVector_0.50.0
## [46] vctrs_0.7.3          Matrix_1.7-4          hms_1.1.4
## [49] bit64_4.8.0          systemfonts_1.3.2     locfit_1.5-9.12
## [52] limma_3.66.0         annotate_1.88.0       glue_1.8.1
## [55] codetools_0.2-20     stringi_1.8.7         gtable_0.3.6
## [58] pillar_1.11.1        htmltools_0.5.9       R6_2.6.1
## [61] textshaping_1.0.5    evaluate_1.0.5        lattice_0.22-7
## [64] png_0.1-9            memoise_2.0.1         Rcpp_1.1.1-1.1
## [67] SparseArray_1.10.10  xfun_0.56             pkgconfig_2.0.3

```